

Ryan McKenna

Previously SC Cleared (eligible for reactivation)

ryan.mckenna002@gmail.com | github.com/RyanPaulMcKenna | [linkedin.com/in/ryan-mckenna-870847165](https://www.linkedin.com/in/ryan-mckenna-870847165) | mckenna-technologies.com | +44 7462 881123 | UK time (Europe/London)

PROFESSIONAL SUMMARY

Embedded Software Engineer and PhD researcher (University of York) specialising in **Embedded Linux, firmware, and low-level systems** with advanced capability in **robotics, physics simulation, and AI control**. Experienced designing and programming complete embedded platforms—from PCB design and bring-up, to Linux kernel drivers and U-Boot, to FPGA/RTOS integration. Research work focuses on physics-based simulation and reinforcement learning for robotic control, bridging **embedded hardware** and **intelligent autonomy**. Skilled at delivering production-grade embedded systems for industrial and defence applications, and developing simulation frameworks that accelerate control-system R&D.

CORE SKILLS

Domains	Embedded Systems • Robotics • Simulation • AI Control
Systems	Embedded C/C++, RTOS, Embedded Linux, Linux kernel drivers, Device Tree, Yocto, MMIO/virtual memory
Hardware	SPI, I ² C, UART, CAN; Oscilloscope/logic analysis; Board bring-up; PCB prototyping
Acceleration	FPGA (VHDL); hardware/software co-design for real-time subsystems
Robotics/ML	Reinforcement Learning, OpenGL, SOFA, ROS1/2, PyTorch; Computer Vision; Control fundamentals
Compliance	RoHS, CE, working knowledge of IEC 61508
Tooling	CMake, GCC/Clang, Git; Python, NumPy/Pandas/Matplotlib; MATLAB

EXPERIENCE

Embedded Systems Software Developer – Linux Printing Technologies, Cambridge Dec 2024 – Present

- Designed and maintained custom Linux kernel device drivers for printer subsystems, interfacing over SPI, I²C, and UART; exposure to PCIe and high-speed interfaces.
- Implemented platform and non-platform drivers; authored device trees; optimised virtual memory mappings for embedded subsystems.
- Built and customised Yocto-based Embedded Linux distributions; configured MMIO and board bring-up for real-time reliability.
- Delivered production firmware in Embedded C/C++ across Embedded Linux and RTOS subsystems with deterministic timing and safe concurrency.
- Designed and built custom test rigs and performed electrical validation (oscilloscope, function generator, thermal imaging) to accelerate debug and ensure RoHS/CE/IEC 61508 compliance.
- Developed and tested FPGA logic (VHDL) integrated with Linux drivers to enhance real-time performance.

Embedded Systems Software Developer – MBDA, Bristol Feb 2022 – Apr 2023

- Developed Embedded Linux and RTOS firmware in MISRA C/C++ for ground-based air defence platforms and radar subsystems.
- Implemented and hardened CAN, Ethernet, and serial communication for distributed embedded nodes; improved reliability and latency.
- Created and modified kernel modules, device trees, and platform drivers for custom defence hardware; integrated board support into Yocto.
- Performed HW/SW bring-up and validation; collaborated with electronics and systems teams under defence constraints; operated under UK SC clearance.

Software Engineer – Elixir Software, Manchester Jul 2019 – Mar 2021

- Built high-performance scientific computing and data analysis tools in C++/Python, boosting lab efficiency and accelerating discovery.
- Applied software engineering best practices (clean reusable code, statistical modelling, visualization) for reliable research systems.

KEY TECHNICAL PROJECTS

Full-Stack Embedded Platform Development

Linux / Personal

Designed and built custom embedded systems end-to-end: schematic design, PCB assembly, firmware development, U-Boot, and Linux kernel driver implementation. Integrated FPGA (VHDL) and RTOS components for deterministic real-time performance.

Physics Simulation & Reinforcement Learning Framework

PhD, University of York

Developed high-fidelity physics simulations (OpenGL/CUDA) for robotic manipulation and control. Implemented physically-informed reinforcement learning and domain-transfer pipelines for sim-to-real robotics and sensor fusion.

Embedded Radar & Communications Systems

MBDA

Developed embedded firmware and low-level drivers for radar platform subsystems in MISRA C/C++. Implemented and hardened CAN, Ethernet, and serial communication for distributed nodes with deterministic scheduling and fault handling. Contributed to bring-up, validation, and integration of real-time data paths within defence constraints.

EDUCATION

2023 – Present **PhD, Computer Science** – University of York (expected completion 2027) (*Robotics, RL, deformable object manipulation*)

Additional Education: MSc Mathematics – University of Chester; BSc Software Engineering – Sheffield Hallam University